

Analysis of .tdb file created (slotted pages)

FROM DBMS MINOR PROJECT (TOYDB) OBJECTIVE 2

How conversion will look like :

```
toydb > splayer > testdata.txt
1 00001001;abhas@aero.iitb.ac.in;;
2 00001002;jain0ua@ccs.iitb.ac.in;;
3 00001003;naik0ua@ccs.iitb.ac.in;;
4 00001004;shyam0ua@ccs.iitb.ac.in;;
5 00001005;pradeep@aero.iitb.ac.in;;

~/Desktop/DBMS_Min_Project/toydb/splayer > hexdump testdata.tdb -C
00000000 ff ff ff ff 01 00 00 00 fe ff ff ff 05 00 00 00 |.....|
00000010 5a 0f 00 00 e0 0f 00 00 20 00 00 00 bf 0f 00 00 |Z.....|
00000020 21 00 00 00 9e 0f 00 00 21 00 00 00 7c 0f 00 00 |!.....|
00000030 22 00 00 00 5a 0f 00 00 22 00 00 00 00 00 00 00 |"...Z..."|
00000040 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
*
00000f60 00 00 00 00 00 00 30 30 30 30 31 30 30 35 3b 70 |.....00001005;p|
00000f70 72 61 64 65 65 70 40 61 65 72 6f 2e 69 69 74 62 |radeep@aero.iitb|
00000f80 2e 61 63 2e 69 6e 3b 3b 30 30 30 30 31 30 30 34 |.ac.in;;00001004|
00000f90 3b 73 68 79 61 6d 30 75 61 40 63 63 73 2e 69 69 |;shyam0ua@ccs.ii|
00000fa0 74 62 2e 61 63 2e 69 6e 3b 3b 30 30 30 30 31 30 |tb.ac.in;;000010|
00000fb0 30 33 3b 6e 61 69 6b 30 75 61 40 63 63 73 2e 69 |03;naik0ua@ccs.i|
00000fc0 69 74 62 2e 61 63 2e 69 6e 3b 3b 30 30 30 30 31 |itb.ac.in;;00001|
00000fd0 30 30 32 3b 6a 61 69 6e 30 75 61 40 63 63 73 2e |002;jain0ua@ccs.|
00000fe0 69 69 74 62 2e 61 63 2e 69 6e 3b 3b 30 30 30 30 |iitb.ac.in;;0000|
00000ff0 31 30 30 31 3b 61 62 68 61 73 40 61 65 72 6f 2e |1001;abhas@aero.|
00001000 69 69 74 62 2e 61 63 2e 69 6e 3b 3b |iitb.ac.in;;|
0000100c
```

This is conversion of testdata.txt having 5 records from .txt to .tdb.

For this doc we will have testdata.txt having 400 records.

Creating testdata.tdb from testdata.txt

```
shreekar@shreeLap ~$ main =
~/Desktop/DBMS_Min_Project/toydb/splayer > ls
build include Makefile src testdata.txt
shreekar@shreeLap ~$ main =
~/Desktop/DBMS_Min_Project/toydb/splayer > make testconvert
build/testconvert testdata.txt 20 0
--- Initializing PF Layer ---
Input file: testdata.txt
Output file: testdata.tdb
Buffer size: 20 pages
Strategy: LRU
-----
--- Conversion Complete ---
Total Records Inserted: 400
Total Pages Used: 5
-----
--- Buffer Manager Statistics ---
Logical Reads: 0
Logical Writes: 5
Physical Reads: 0
Physical Writes: 5
Buffer Hits: 0
Hit Rate: -nan%
---
shreekar@shreeLap ~$ main =
~/Desktop/DBMS_Min_Project/toydb/splayer > ls
build include Makefile src testdata.tdb testdata.txt
shreekar@shreeLap ~$ main =
~/Desktop/DBMS_Min_Project/toydb/splayer >
```

Converted testdata.txt (having 400 records) into testdata.tdb file (having 5 slotted pages)

On the next page I discussed the structure of testdata.tdb.

Analysis of testdata.tdb :

1. First 4 bytes of testdata.tdb :

```
shreekar@shreeLap ~$ main =
~/Desktop/DBMS_Min_Project/toydb/splayer > hexdump testdata.tdb -C
00000000 ff ff ff ff 05 00 00 00 fe ff ff ff 62 00 00 00 |.....b...|
00000010 27 03 00 00 e0 0f 00 00 20 00 00 00 bf 0f 00 00 |'.....|
00000020 21 00 00 00 9e 0f 00 00 21 00 00 00 7c 0f 00 00 |!.....!...|...|
00000030 22 00 00 00 5a 0f 00 00 22 00 00 00 3c 0f 00 00 |"...Z..."...<...|
00000040 1e 00 00 00 1a 0f 00 00 22 00 00 00 f8 0e 00 00 |.....".....|
00000050 22 00 00 00 d7 0e 00 00 21 00 00 00 b5 0e 00 00 |".....!.....|
00000060 22 00 00 00 95 0e 00 00 20 00 00 00 73 0e 00 00 |".....s...|
00000070 22 00 00 00 51 0e 00 00 22 00 00 00 31 0e 00 00 |"...Q..."...1...|
00000080 20 00 00 00 0f 0e 00 00 22 00 00 00 ed 0d 00 00 |"...|
```

```
First word : ff ff ff ff = 0xffffffff = -1 (signed INT)
Second word : 05 00 00 00 = 0x00000005 = 5 (signed INT)
Third word : fe ff ff ff = 0xfffffffffe = -2 (signed INT)
Fourth word : 62 00 00 00 = 0x00000062 = 98 (signed INT)
```

1. First word : Pointing to first Free page in testdata.tdb
2. Second word : Number of pages in testdata.tdb
Now first page started :
3. Third word : -2 (page in use), -1 (then page is not in use and it is last free page), +ve int (then page is free but not last free page, so it is pointing to next free page, creating linked list of free pages)
4. Fourth word : Number of record in that (first page) page

2. Starting of Second Page in testdata.tdb file :

```
00000fe0 69 69 74 62 2e 61 63 2e 69 6e 3b 3b 30 30 30 30 |iitb.ac.in;;0000|
00000ff0 31 30 30 31 3b 61 62 68 61 73 40 61 65 72 6f 2e |1001;abhas@aero.|
00001000 69 69 74 62 2e 61 63 2e 69 6e 3b 3b fe ff ff ff |iitb.ac.in;;....|
00001010 63 00 00 00 41 03 00 00 de 0f 00 00 22 00 00 00 |c...A....."....|
00001020 bc 0f 00 00 22 00 00 00 9a 0f 00 00 22 00 00 00 |....."....."....|
00001030 78 0f 00 00 22 00 00 00 54 0f 00 00 24 00 00 00 |x..."...T...$...|
00001040 31 0f 00 00 23 00 00 00 0f 0f 00 00 22 00 00 00 |1...#....."....|
00001050 f0 0e 00 00 1f 00 00 00 ce 0e 00 00 22 00 00 00 |....."....."....|
00001060 ae 0e 00 00 20 00 00 00 8c 0e 00 00 22 00 00 00 |....."....."....|
00001070 6a 0e 00 00 22 00 00 00 4a 0e 00 00 20 00 00 00 |j..."...J... ..|
00001080 28 0e 00 00 22 00 00 00 08 0e 00 00 20 00 00 00 |(..."....."....|
```

Highlighted word is first word for 2nd page (which is 0xfffffffffe = -2(decimal), showing that page is not free)

Next word to highlighted word is 63 00 00 00 = 0x00000063 = 99(decimal), showing number of records in that (second page) page